Multimodal Fake News & Image Tampering Diagnostic Report

Generated on 2026-08-02 16:37:37 IST | Project Forensic Audit
Analysis performed at 2026-08-02 16:34:48 IST

1. Executive Summary & Verdict

Final Classification Verdict	**REAL**
Overall Multimodal Risk Score	0.4540 (0.0=Real, 1.0=Fake)
Prediction Confidence	57.26%
Processing Time	1.57 seconds

This submission was classified as likely authentic, with limited confidence (57.3%). The article's text contributed most to this verdict (47% of the fused score), primarily because attribution to a credible source found: according to. Additional supporting evidence: text model attribution flagged these tokens as most fake-indicative: renewable energy.

2. Evidence Summary

• Attribution to a credible source found: according to.
• Text model attribution flagged these tokens as most fake-indicative: renewable energy.
• Error Level Analysis found no strong splicing/re-compression signal (anomaly score 0.162).
• Image classifier assigned this image an elevated manipulation-risk score.
• Histogram analysis found hard tonal clipping, consistent with aggressive brightness/contrast editing.
• No EXIF metadata was found in the image file.
• Simulated propagation graph shows an elevated proportion of suspicious/bot-like spreader accounts (16%).

3. Model Results & Fusion Weights

Modality	Model Engine	Risk Score	Weight	Relative Impact
Text Module	TF-IDF / BERT Transformer	0.5348	0.40	47.1%
Image Module	ResNet50 / ELA / pHash	0.5173	0.35	39.9%
Graph Module	PyG Graph Attention (GAT)	0.2364	0.25	13.0%

4. Analyzed Article Content

"The Ministry of New and Renewable Energy (MNRE) has announced the commissioning of a new 300 MW solar power project in Rajasthan. According to the official press release, the project is expected to increase renewable energy capacity while reducing carbon emissions. Engineers completed performance and safety testing before connecting the plant to the national power grid. image of Solar farm"

5. Text Explanation — Top Contributing Tokens

SHAP/coefficient attribution identified the following tokens as most influential in the text-model verdict: **official, according, project, solar, announced, carbon, carbon emissions, emissions**.

6. Image Explanation

Error Level Analysis showed no strong splicing signal (anomaly score 0.1624). Perceptual hash: `a90082a040008000`. A GradCAM attention heatmap localizing the regions most influential to the ResNet-50 verdict was generated for this image (see dashboard for the visual overlay).

Forensic observations: no EXIF metadata was found (common after re-saving, screenshotting, or social-media re-upload, but also consistent with metadata having been deliberately stripped); hard tonal clipping was detected in the R, B channel(s), consistent with aggressive brightness/contrast editing.

7. Propagation Analysis Summary

Simulated propagation graph: 45 nodes, 96 cascade edges, 6 detected communities, diffusion depth 5, 16% suspicious/bot-like accounts. Graph Attention Network propagation risk score: 0.2364.

8. Fusion Analysis

The final verdict is a weighted linear combination of the unimodal risk scores (see Section 3), calibrated via Platt scaling fit on the project's labeled demo corpus. Any modality not submitted for this analysis defaults to a neutral 0.5 score rather than being fabricated. The modality with the largest 'Relative Impact' in the table above contributed most to the final classification.

9. Final Decision & Recommendations

This content does not show strong indicators of fabrication based on the signals available to this system. As with any automated assessment, cross-verification against primary sources is still recommended before using this content for high-stakes decisions.